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Preview

Candidozyma auris turns skin immunity to its advantage

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https://doi.org/10.1016/j.chom.2026.09.004

Skin colonization is central to *Candidozyma auris* success as a healthcare-associated pathogen. In a recent *Science* paper, Merrill et al. show that during skin colonization, *C. auris* occupies hair follicles and exposes chitin, which is associated with induction of IFNγ responses that remodel follicular keratinocytes to promote fungal persistence.

Unlike most pathogenic *Candida* species, *Candidozyma auris* has a remarkable ability to persist on human skin. This persistent colonization creates a reservoir for transmission and increases the risk of invasive infection. Emerging studies suggest that successful skin colonization requires several coordinated processes, including efficient adhesion to skin surfaces, physiological adaptation to the nutrient-poor and stressful skin environment, and active modulation of host immunity.

The first challenge for a skin colonizer is to establish stable attachment. Hair follicles provide a distinct niche on the skin, and recent work by Merrill et al. uses a combination of electron microscopy of human hair colonized with *C. auris* and hairless mouse models to show that *C. auris* can directly bind to hair.[1] This complements recent genetic studies to identify fungal factors that enable *C. auris* to attach to and colonize skin. Santana et al. identified Scf1, a *C. auris*-specific adhesin together with a conserved adhesin Iff4109, that promotes surface attachment, biofilm formation, and skin colonization.[2] Another distinct adhesion mechanism is mediated by Als4112. Zhao et al. showed that this conserved cell-surface adhesin promotes skin colonization through attachment to keratinocytes and extracellular matrix proteins such as laminin.[3] Together, these findings establish adhesion as an important first step in *C. auris* host colonization.

Adhesion alone, however, cannot explain the ability of *C. auris* to persist on skin for months. Human skin is nutrient poor, acidic, osmotically stressful, and chemically distinct from conventional laboratory media. Nicklas and colleagues found that *C. auris* isolates from multiple clades grow more efficiently than *Candida albicans* in artificial media that mimic nutrients available on skin and undergo extensive metabolic reprogramming under these conditions.[4] Disrupting genes involved in alternative carbon metabolism impaired growth in skin-like media, highlighting the importance of metabolic flexibility to continued skin persistence. To complement these findings, Glawe and colleagues identified potassium homeostasis through the Trk1 transporter as important for colonization.[5] Thus, *C. auris* appears to combine attachment with metabolic and stress-adaptation programs that allow it to survive the challenging skin environment.

Fungal attachment and physiological adaptation may still be only part of the story. *C. auris* isolates from different clades show a significant variation in their ability to colonize skin in an *in vivo* murine model[6] and *in vitro* human keratinocyte models[3] but show relatively little variation in *ex vivo* human or porcine skin models.[7] This difference raises the possibility that components missing from *ex vivo* systems, including host immunity, help determine colonization in the intact human host. The emerging picture is therefore one in which fungal attachment establishes adherence to the skin, physiological adaptation supports persistence, and interactions with the host further shape whether colonization succeeds.

Merrill et al. reveal a striking link between fungal localization and the host immune response.[1] Whereas *C. albicans* induces protective type 17 immunity and is efficiently cleared from skin, *C. auris* persists within hair follicles and elicits an IFNγ response that is non-protective. Dendritic cells and IFNγ-producing T cells accumulate around colonized follicles, where IFNγ acts on keratinocytes to weaken barrier function and antifungal defenses, favoring fungal persistence. This builds on the work from Das et al., who demonstrated that excessive IFNγ increases *C. auris* burden while suppressing protective IL-17 responses, and the work from Break et al., who demonstrated a pathogenic role for IFNγ in *Candida albicans* infections.[8,9] Notably, Das et al. identified the outer mannan layer as the fungal stimulus that drives the IFNγ response through IL-12-producing inflammatory myeloid cells.[8] Merrill et al. add to this picture in which fungal cell wall components influence skin immune responses by showing that increased chitin exposure is associated with an increased IFNγ response.[1] They demonstrated that heat-killed *C. auris* retained the ability to induce IFNγ, indicating that an active fungal process was not required. Cell wall staining pointed to chitin as a potential inducing signal. Furthermore, experimentally altering cell wall architecture through treatment with the Fks1 inhibitor caspofungin, which drives a compensatory increase in chitin exposure, changed the magnitude of the IFNγ response. Additionally, a mutant with reduced chitin exposure elicited less IFNγ and showed reduced fungal burden on skin. Together, these studies suggest that *C. auris* cell-surface architecture helps shape skin immunity to promote fungal persistence. The distinctive skin immune response to *C. auris* may be determined not by a single fungal molecule but by the overall organization of its cell surface. Mannan forms the outer layer of the fungal cell wall and can influence exposure of underlying components, whereas chitin is

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normally buried more deeply. The involvement of both mannan and chitin therefore suggests that coordinated cell-wall organization and ligand exposure may shape immune recognition. The upstream signals that trigger chitin exposure in *C. auris* on the skin, as well as the host cells and receptors that sense this change, remain to be defined. This is complicated by the fact that the chitin polysaccharide obtained from different sources can have different structural properties, making it difficult to analyze.

The skin environment itself may provide part of the answer. On one hand, Merrill et al. demonstrated that skin-like synthetic sweat medium induced chitin exposure, which then is associated with increased IFNγ responses.[1] On the other hand, *C. auris* encounters not only host-derived nutrients and stresses, but also a complex resident microbiome. Clinical metagenomic studies have found distinct microbial communities in *C. auris*-colonized and non-colonized individuals, although whether these differences contribute to colonization remains unclear. Proctor et al., for example, observed correlation between multidrug-resistant bacterial species and *C. auris* colonization, but anticorrelation between *Malassezia* and *C. auris*.[10] During skin colonization, it is likely that a combination of skin nutrients, environmental stresses, and signals from neighboring microbes reshape the *C. auris* cell wall, thereby influencing both fungal fitness and host recognition. Determining how the skin environment controls fungal cell wall architecture and pathogen-associated molecular pattern (PAMP) exposure may therefore provide the missing link between adaptation to the skin niche and the immune response elicited by *C. auris*.

C. auris is unusual for a fungal pathogen in that it is only recently associated with humans, but all isolates still appear to be human-associated or closely related to human-associated environments. The true environmental niche and reservoir for this organism is still undefined, and the adaptations and virulence factors that allowed it to be a successful human pathogen are still being discovered. It is likely that adaptation to this currently unknown environmental niche provided the training ground that allowed *C. auris* to become a successful human pathogen, and that discovery of that niche will provide invaluable insight into the selective pressures on *C. auris*.

Due to the recent association with humans, it is unlikely that there has been significant co-evolution with mammalian hosts, which brings up many questions about how *C. auris* has adapted to human immune responses, or how the host responds to *C. auris* challenge. A recent focus in the field has been on identifying these host-microbe interactions and adaptations, with the idea that these virulence factors can then be exploited as new potential antifungal drug targets to address the challenge of multidrug-resistant fungi. In the work from Merrill et al., identifying how the host responds to *C. auris* is a critical insight into how that host-microbe balance can be leveraged to treat fungal colonization. Potentially, IFNγ inhibitors may be useful to skew the host immune response to improve barrier integrity and prevent *C. auris* colonization of the hair follicle.

Together, these studies suggest that persistent *C. auris* colonization emerges from several cooperating fungal traits: adhesins establish contact, stress and metabolic programs support survival and growth, the surrounding microbiome may shape local fitness, and dynamic remodeling of the fungal surface influences host recognition. Merrill et al. add a new dimension by showing that the immune response elicited by these fungal adaptations can, in turn, reshape the skin niche to favor fungal persistence. As chitin is a common component of fungal cell walls, there are still many open questions about the role of chitin in determining skin immunity, and the balance between specific pathogen-associated molecular pattern exposure and host immune responses. It is possible that the immune response to chitin is normally advantageous in protecting the host from other fungal pathogens that do not have the same attachment or persistence potential as *C. auris.* Understanding how these programs work together may reveal new strategies to prevent persistent *C. auris* colonization.

ACKNOWLEDGMENTS

T.R.O. was supported by NIH grant U19AI181767 and grant 1173374 from the Burroughs Wellcome Fund. G.Z. was supported by a fellowship from the Michigan Postdoctoral Pioneers.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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